

Physical Chemistry–III (CHB-504)

Unit 3: Surface Chemistry & Enzyme Kinetics

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Don't cry - Padhna hi padega

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Adsorption: Introduction and Types

Adsorption vs. Absorption

Feature	Adsorption	Absorption
Definition	Accumulation of a substance on a surface (2-D phenomenon)	Uniform distribution of a substance throughout a bulk phase (3-D)
Location	Surface only	Entire bulk
Example	N ₂ on charcoal	Water in sponge
Reversibility	Often reversible	Often irreversible

Sorption = combined adsorption + absorption when both occur.

Physisorption vs. Chemisorption

Feature	Physisorption	Chemisorption
Force	Van der Waals (weak)	Chemical bond (covalent/ionic)
ΔH_{ads}	Low: 5–40 kJ/mol	High: 40–400 kJ/mol
Reversibility	Easily reversible	Often irreversible
Temperature	Favoured at low T	Increases initially, then decreases
Layers	Multilayer possible	Monolayer only
Specificity	Non-specific	Highly specific
Activation energy	None (no barrier)	Often requires activation energy
Example	N ₂ on silica at 77 K	H ₂ on Ni, CO on Pt

Why Adsorption is Spontaneous

For adsorption: $\Delta G_{\text{ads}} < 0$ (spontaneous). Since the adsorbed molecule loses translational freedom:

$$\Delta S_{\text{ads}} < 0$$

For spontaneity: $\Delta G = \Delta H - T\Delta S < 0$ requires $\Delta H_{\text{ads}} < 0$ (exothermic). Thus adsorption is always **exothermic**.

This also explains why adsorption *decreases* with increasing temperature (Le Chatelier's principle).

Why Surface Reactions Have Lower Activation Energy

On a surface, the adsorbed reactant molecules are:

- Oriented favourably for reaction (reduced entropy penalty).
- Weakened in bonding (partial activation already provided by adsorption energy).
- Concentrated in a small region (effectively higher local concentration).

Hence the activation energy $E_a^{\text{surface}} < E_a^{\text{homogeneous}}$.

Langmuir Adsorption Isotherm

Assumptions

1. The surface has a fixed, finite number of adsorption sites, all **equivalent**.
2. Each site can hold **at most one** molecule (monolayer adsorption).
3. Adsorbed molecules do **not** interact with each other.
4. Adsorption is a **dynamic equilibrium** between adsorption and desorption.
5. The surface is **energetically homogeneous**.

Derivation of the Langmuir Isotherm

Let θ = fraction of surface covered (fractional coverage), so $(1 - \theta)$ = fraction bare.

Rate of adsorption $\propto$ pressure P and fraction of bare surface:

$$r_{\text{ads}} = k_a P(1 - \theta)$$

Rate of desorption $\propto$ fraction of covered surface:

$$r_{\text{des}} = k_d \theta$$

At **dynamic equilibrium**: $r_{\text{ads}} = r_{\text{des}}$:

$$k_a P(1 - \theta) = k_d \theta$$

$$k_a P - k_a P \theta = k_d \theta$$

$$k_a P = \theta(k_d + k_a P)$$

$$\boxed{\theta = \frac{KP}{1 + KP}}$$

where $K = k_a/k_d$ is the **Langmuir adsorption constant** (equilibrium constant for adsorption, units: pressure⁻¹).

Limiting Cases

1. **Low pressure** ($KP \ll 1$): $1 + KP \approx 1$, so:

$$\theta \approx KP \Rightarrow \theta \propto P \quad (\text{first-order, Henry's law region})$$

2. **High pressure** ($KP \gg 1$): $1 + KP \approx KP$, so:

$$\theta \approx 1 \quad (\text{surface fully covered, zero-order in } P)$$

3. **Intermediate pressure**: $\theta = KP/(1 + KP)$ (Langmuir form).

Langmuir Isotherm Applied to X(g) Adsorption on Metal

The $X_2(g)$ adsorption with dissociation on a metal surface follows:



where S is a surface site. Rate of adsorption $\propto P$ and $(1 - \theta)^2$; rate of desorption $\propto \theta^2$:

$$k_a P (1 - \theta)^2 = k_d \theta^2$$

$$\frac{\theta^2}{(1 - \theta)^2} = \frac{k_a P}{k_d} = KP$$

$$\boxed{\theta = \frac{(KP)^{1/2}}{1 + (KP)^{1/2}}}$$

Ratio of adsorption to desorption at 1 atm: Given fractional coverage $\theta = 0.50$ at 1 atm, substituting into the isotherm:

$$0.50 = \frac{(K \times 1)^{1/2}}{1 + (K \times 1)^{1/2}} \implies K = 1 \text{ atm}^{-1}$$

Ratio $= \theta/(1 - \theta) = 0.50/0.50 = 1.0$ (equal rates at $\theta = 0.5$, by definition of equilibrium).

Worked Example: Fractional Coverage

Given: $K = 0.3 \text{ kPa}^{-1}$, $P = 0.2 \text{ kPa}$. Find θ .

$$\theta = \frac{KP}{1 + KP} = \frac{0.3 \times 0.2}{1 + 0.3 \times 0.2} = \frac{0.06}{1.06} = \boxed{0.0566}$$

BET (Brunauer–Emmett–Teller) Theory**Need for BET Theory**

The Langmuir isotherm assumes **monolayer** adsorption. In practice (especially physisorption), **multilayer** adsorption occurs. The BET theory extends Langmuir to multilayers.

Assumptions of BET Theory

1. Gas molecules can adsorb in **multiple layers** on a surface.
2. The first layer has a heat of adsorption ΔH_1 (specific to the surface).
3. All subsequent layers have heat of adsorption equal to the heat of liquefaction ΔH_L .
4. Langmuir equilibrium applies to each layer independently.
5. The number of layers $\rightarrow \infty$ at saturation pressure P^0 (when the gas condenses).
6. There are **no** lateral interactions between adsorbed molecules in the same layer.

The BET Equation

$$\frac{P}{V(P^0 - P)} = \frac{1}{V_m c} + \frac{(c - 1)}{V_m c} \cdot \frac{P}{P^0}$$

where:

- P = pressure of adsorbate gas
- P^0 = saturation (vapour) pressure of adsorbate at that temperature
- V = volume of gas adsorbed at pressure P (measured at STP)
- V_m = volume of gas required to form a **complete monolayer** (at STP)
- c = BET constant related to adsorption enthalpy: $c \approx \exp[(\Delta H_1 - \Delta H_L)/RT]$

Linear Form and How to Use BET to Find Surface Area

Plot $\frac{P}{V(P^0 - P)}$ vs. $\frac{P}{P^0}$: gives a straight line.

$$\text{Slope} = \frac{c - 1}{V_m c}, \quad \text{Intercept} = \frac{1}{V_m c}$$

From slope and intercept:

$$V_m = \frac{1}{\text{slope} + \text{intercept}}, \quad c = \frac{\text{slope}}{\text{intercept}} + 1$$

Surface area calculation:

$$S_{\text{BET}} = \frac{V_m}{22400} \times N_A \times a_m$$

where $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$ and a_m is the cross-sectional area of one adsorbate molecule (for N_2 : $a_m = 16.2 \text{ \AA}^2 = 16.2 \times 10^{-20} \text{ m}^2$).

Worked Example: Surface Area of Silica Gel

Given: $V_m = 610$ cc of N_2 (at STP) covers 5 g of silica gel. Density of N_2 at STP = 0.03 g/cc.

Step 1: Moles of N_2 in monolayer:

$$n = \frac{V_m}{22400 \text{ cc/mol}} = \frac{610}{22400} = 0.02723 \text{ mol}$$

Step 2: Number of molecules:

$$N = n \times N_A = 0.02723 \times 6.022 \times 10^{23} = 1.639 \times 10^{22} \text{ molecules}$$

Step 3: Surface area (using $a_m = 16.2 \times 10^{-20} \text{ m}^2$ for N_2):

$$S = N \times a_m = 1.639 \times 10^{22} \times 16.2 \times 10^{-20} = 2655 \text{ m}^2$$

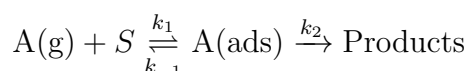
Specific surface area = $2655/5 = \boxed{531 \text{ m}^2 \text{ g}^{-1}}$.

Why Langmuir Describes Chemisorption Better than Physisorption

- **Chemisorption** is monolayer (one molecule per site, strong bond), so the Langmuir monolayer assumption is valid.
- **Physisorption** involves multilayer formation (weak Van der Waals forces allow stacking), which Langmuir cannot account for – BET is needed.

Unimolecular Surface Reactions (Langmuir–Hinshelwood)**Mechanism**

A reactant A adsorbs on the surface and then reacts (decomposes/rearranges):

**Rate Expression**

The rate of reaction = $k_2 \theta_A \times N_s$ (where N_s = total surface sites per unit area):

$$r = k_2 \theta_A = k_2 \frac{K_A P_A}{1 + K_A P_A}$$

Limiting Cases and Order of Reaction

1. **Low pressure** ($K_A P_A \ll 1$):

$$r \approx k_2 K_A P_A \propto P_A \Rightarrow \text{First-order}$$

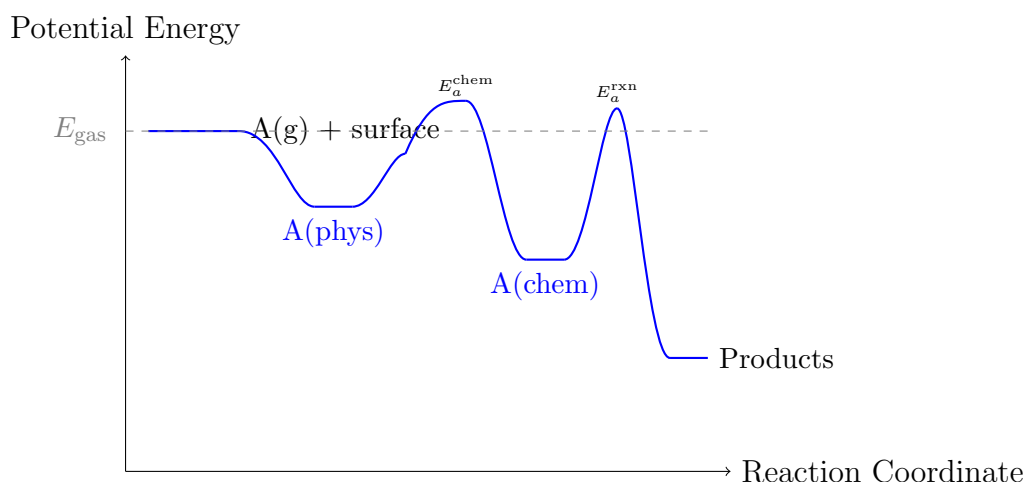
The surface is sparsely covered; more reactant increases rate linearly.

2. **High pressure** ($K_A P_A \gg 1$):

$$r \approx k_2 \quad (\text{independent of } P_A) \Rightarrow \text{Zero-order}$$

Surface is fully covered (saturated); rate is limited by k_2 alone.

Potential Energy Diagram for Unimolecular Surface Reaction



At low pressure: Surface sparsely covered; reaction is first-order (follows $\theta \approx KP$, rate $\propto P$). Activation energy = $E_a^{\text{chem}} + E_a^{\text{rxn}}$.

At high pressure: Surface saturated; rate = k_2 , zero-order. The apparent E_a is just E_a^{rxn} (chemisorption step already completed).

Effect of Inhibitor on Unimolecular Surface Reaction

Competitive Inhibitor

An inhibitor I competes with reactant A for the same surface sites:

$$\theta_A = \frac{K_A P_A}{1 + K_A P_A + K_I P_I}, \quad \theta_I = \frac{K_I P_I}{1 + K_A P_A + K_I P_I}$$

Rate:

$$r = k_2 \theta_A = \frac{k_2 K_A P_A}{1 + K_A P_A + K_I P_I}$$

Case: Strongly Adsorbed Inhibitor, Low Reactant Pressure

When $K_I P_I \gg 1$ and $K_I P_I \gg K_A P_A$ (inhibitor dominates surface):

$$r \approx \frac{k_2 K_A P_A}{K_I P_I} \propto \frac{P_A}{P_I}$$

Effect on reaction order:

- At low P_A with **strongly adsorbed inhibitor**: $r \propto P_A/P_I$ – still first order in A, but **inverse first-order in inhibitor**.
- The effective activation energy increases: $E_a^{\text{apparent}} = E_a^{\text{reaction}} - \Delta H_{\text{ads}}(I) + \Delta H_{\text{des}}(I)$, generally larger.

Potential Energy Diagram with Inhibitor

The inhibitor raises the effective activation energy:

- Without inhibitor: Reactant A occupies sites easily, E_a = intrinsic barrier.
- With strong inhibitor: A must displace I from the surface, requiring extra energy = $\Delta H_{\text{ads}}(I)$, so apparent E_a is *larger*.

Decomposition of ammonia on platinum is inhibited by H_2 (strongly adsorbed). When the surface is sparsely covered by NH_3 but fully covered by H_2 :

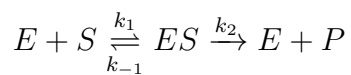
$$r = \frac{k_2 K_{\text{NH}_3} P_{\text{NH}_3}}{K_{\text{H}_2} P_{\text{H}_2}}$$

Apparent activation energy = $E_a(\text{decomp}) + \Delta H_{\text{des}}(\text{H}_2) > E_a(\text{without inhibitor})$.

Enzyme Catalysis and Michaelis–Menten Kinetics

Introduction

Enzymes are biological catalysts (proteins) that dramatically accelerate reactions. The substrate (S) binds to the active site of the enzyme (E) to form an enzyme–substrate complex (ES), which then yields product (P):



Derivation of Michaelis–Menten Equation

Assumptions:

1. The total enzyme concentration $[E]_0 = [E] + [ES]$ is conserved.
2. The substrate concentration $[S] \gg [E]_0$ (enzyme is not saturated immediately).
3. **Steady-state approximation:** The concentration of ES reaches a steady state quickly:

$$\frac{d[ES]}{dt} = 0$$

Rate equations:

$$\frac{d[ES]}{dt} = k_1[E][S] - k_{-1}[ES] - k_2[ES] = 0$$

$$k_1[E][S] = (k_{-1} + k_2)[ES]$$

From conservation: $[E] = [E]_0 - [ES]$:

$$k_1([E]_0 - [ES])[S] = (k_{-1} + k_2)[ES]$$

$$k_1[E]_0[S] = [ES](k_{-1} + k_2 + k_1[S])$$

$$[ES] = \frac{k_1[E]_0[S]}{k_{-1} + k_2 + k_1[S]} = \frac{[E]_0[S]}{K_M + [S]}$$

where the **Michaelis constant** is:

$$K_M = \frac{k_{-1} + k_2}{k_1}$$

Rate of product formation:

$$v = k_2[ES] = \frac{k_2[E]_0[S]}{K_M + [S]}$$

Maximum rate $V_{\max} = k_2[E]_0$ (when all enzyme is in ES form):

$$v = \frac{V_{\max}[S]}{K_M + [S]}$$

This is the **Michaelis–Menten equation**.

Physical Significance of K_M

K_M is the substrate concentration at which the reaction rate is **half the maximum rate**:

$$v = \frac{V_{\max}}{2} \text{ when } [S] = K_M$$

Proof: Set $v = V_{\max}/2$ in the Michaelis–Menten equation:

$$\frac{V_{\max}}{2} = \frac{V_{\max}[S]}{K_M + [S]} \implies K_M + [S] = 2[S] \implies [S] = K_M$$

K_M has units of concentration (mol/L). A **low** K_M indicates high affinity of enzyme for substrate (forms ES complex readily). K_M is a property of the specific enzyme–substrate pair.

Limiting Cases of Michaelis–Menten Equation

1. $[S] \ll K_M$: $K_M + [S] \approx K_M$:

$$v \approx \frac{V_{\max}}{K_M}[S] \implies \textbf{First-order in [S]}$$

Rate increases linearly with substrate concentration.

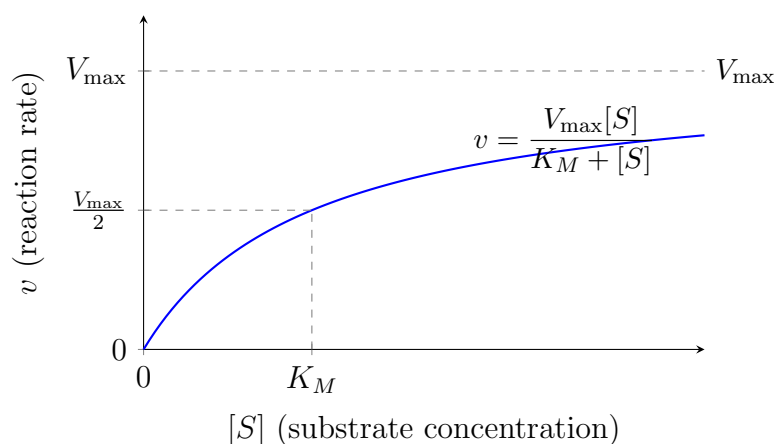
2. $[S] \gg K_M$: $K_M + [S] \approx [S]$:

$$v \approx V_{\max} \implies \textbf{Zero-order in [S]}$$

Rate is independent of substrate concentration; enzyme is saturated.

3. $[S] = K_M$: $v = V_{\max}/2$ (defines K_M).

Rate vs. Substrate Concentration Graph



Effect of pH and Temperature on Enzyme-Catalysed Reactions

Effect of pH

- Each enzyme has an **optimum pH** at which activity is maximum (usually 6–8, e.g., pepsin at pH 2, trypsin at pH 8).
- **Below optimum:** Protonation of active site residues alters enzyme shape or blocks substrate binding.
- **Above optimum:** Deprotonation changes charge distribution; enzyme–substrate interaction weakens.
- **Extreme pH:** Denaturation of enzyme protein destroys activity irreversibly.
- $V_{\max}$ and K_M both change with pH; the bell-shaped v vs. pH curve reflects two ionisation equilibria flanking the active site.

Effect of Temperature

- **Below optimum T :** Rate increases with T following Arrhenius law ($k = Ae^{-E_a/RT}$).
- **At optimum T :** Maximum activity (typically 37°C for human enzymes).
- **Above optimum T :** Denaturation of enzyme protein (irreversible unfolding) rapidly decreases activity.
- The v vs. T curve shows a sharp bell shape with a maximum at the optimum temperature.

Lineweaver–Burk Plot (Double Reciprocal Plot)

Derivation

Taking the reciprocal of both sides of the Michaelis–Menten equation:

$$v = \frac{V_{\max}[S]}{K_M + [S]}$$

$$\frac{1}{v} = \frac{K_M + [S]}{V_{\max}[S]} = \frac{K_M}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

$$\boxed{\frac{1}{v} = \frac{K_M}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}}$$

This is a linear equation: $y = mx + c$ where:

$$y = 1/v$$

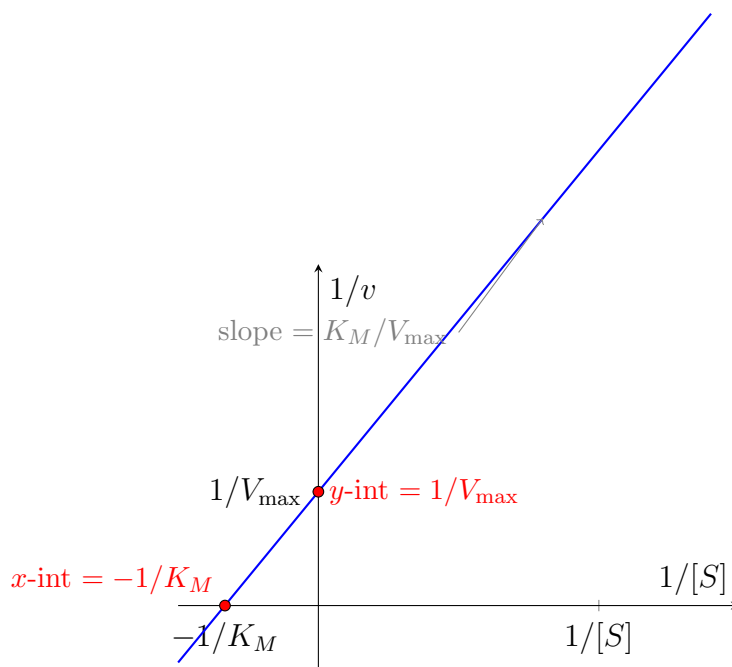
$$x = 1/[S]$$

$$\text{Slope} = K_M/V_{\max}$$

$$\text{Intercept (y-axis, } 1/[S] = 0) = 1/V_{\max}$$

$$\text{Intercept (x-axis, } 1/v = 0) = -1/K_M$$

Lineweaver–Burk Plot Diagram



Advantages and Disadvantages

Advantages:

- Converts hyperbolic MM curve to a **straight line** – easy to determine K_M and $V_{\max}$ graphically.
- Simple linear regression can be used.

Disadvantages:

- Compresses data at high $[S]$ (small $1/[S]$ values), distorting errors.
- Small errors at low v are amplified by taking $1/v$.
- Better alternatives: Eadie–Hofstee plot (v vs. $v/[S]$), Hanes–Woolf plot ($[S]/v$ vs. $[S]$).

Eadie–Hofstee Plot

Rearranging the Michaelis–Menten equation:

$$v(K_M + [S]) = V_{\max}[S]$$

$$vK_M + v[S] = V_{\max}[S]$$

$$v = V_{\max} - K_M \cdot \frac{v}{[S]}$$

$$v = -K_M \left(\frac{v}{[S]} \right) + V_{\max}$$

Plot v (y-axis) vs. $v/[S]$ (x-axis): straight line with slope $= -K_M$, y-intercept $= V_{\max}$, x-intercept $= V_{\max}/K_M$.

Stern–Volmer Equation (Brief, for Context)

Though primarily a photochemistry topic, the Stern–Volmer equation appears in enzyme kinetics context:

$$\frac{v_0}{v} = 1 + K_{SV}[Q]$$

where $[Q]$ is the concentration of quencher/inhibitor and K_{SV} is the Stern–Volmer constant. A plot of v_0/v vs. $[Q]$ gives a straight line with slope K_{SV} .

Worked Problems

Michaelis–Menten Calculation

Problem: Limiting rate $V_{\max} = 1.2 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$, $K_M = 1.2 \times 10^{-4} \text{ mol L}^{-1}$, substrate $[S] = 9.99 \text{ mol L}^{-1}$. Find $[S]$ when $v = 3.6 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$.

Note: Check if $v > V_{\max}$. Here $V_{\max} = 1.2 \times 10^{-4}$ but $v = 3.6 \times 10^{-4} > V_{\max}$. This suggests either the data relates to a different enzyme system or $V_{\max}$ at a higher $[E]_0$. Assuming the problem means to find the rate at $[S] = 9.99$:

$$v = \frac{V_{\max}[S]}{K_M + [S]} = \frac{1.2 \times 10^{-4} \times 9.99}{1.2 \times 10^{-4} + 9.99} \approx \frac{1.2 \times 10^{-4} \times 9.99}{9.990} \approx 1.2 \times 10^{-4} \text{ mol L}^{-1}\text{s}^{-1}$$

Since $[S] \gg K_M$, the rate is essentially $V_{\max}$.

Ionic Strength Calculation

Problem: Calculate the ionic strength of (i) 50 mL of 0.4 M K_2SO_4 + 50 mL of 0.4 M BaCl_2 .

After mixing: total volume = 100 mL, so concentrations halved.

$$[\text{K}^+] = 2 \times 0.2 = 0.4 \text{ M}, \quad z = +1$$

$$[\text{SO}_4^{2-}] = 0.2 \text{ M}, \quad z = -2$$

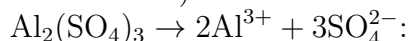
$$[\text{Ba}^{2+}] = 0.2 \text{ M}, \quad z = +2$$

$$[\text{Cl}^-] = 2 \times 0.2 = 0.4 \text{ M}, \quad z = -1$$

$$I = \frac{1}{2} \sum_i c_i z_i^2 = \frac{1}{2} [0.4(1)^2 + 0.2(4) + 0.2(4) + 0.4(1)] = \frac{1}{2} [0.4 + 0.8 + 0.8 + 0.4] = \frac{1}{2} (2.4) = \boxed{1.2 \text{ mol L}^{-1}}$$

Ionic Strength of $\text{Al}_2(\text{SO}_4)_3 + \text{K}_4\text{Fe}(\text{CN})_6$

50 mL of 0.2 M $\text{Al}_2(\text{SO}_4)_3$ + 50 mL of 0.2 M $\text{K}_4\text{Fe}(\text{CN})_6$ (total 100 mL, concentrations halved to 0.1 M):



$$[\text{Al}^{3+}] = 0.2 \text{ M}, z = 3; \quad [\text{SO}_4^{2-}] = 0.3 \text{ M}, z = -2$$



$$[\text{K}^+] = 0.4 \text{ M}, z = 1; \quad [\text{Fe}(\text{CN})_6^{4-}] = 0.1 \text{ M}, z = -4$$

$$I = \frac{1}{2}[0.2(9) + 0.3(4) + 0.4(1) + 0.1(16)] = \frac{1}{2}[1.8 + 1.2 + 0.4 + 1.6] = \frac{1}{2}(5.0) = \boxed{2.5 \text{ mol L}^{-1}}$$

Summary Table

Topic	Key Result
Langmuir isotherm	$\theta = KP/(1 + KP)$
Dissociative adsorption	$\theta = (KP)^{1/2}/(1 + (KP)^{1/2})$
BET equation	$P/[V(P^0 - P)] = 1/(V_m c) + (c - 1)P/(V_m c P^0)$
Surface area (BET)	$S = (V_m/22400) \times N_A \times a_m$
Surface rxn rate	$r = k_2 K_A P_A / (1 + K_A P_A)$
With inhibitor	$r = k_2 K_A P_A / (1 + K_A P_A + K_I P_I)$
Michaelis–Menten	$v = V_{\max}[S]/(K_M + [S])$
K_M significance	$[S]$ at which $v = V_{\max}/2$
Lineweaver–Burk	slope = $K_M/V_{\max}$; y-int = $1/V_{\max}$; x-int = $-1/K_M$
Eadie–Hofstee	$v = V_{\max} - K_M(v/[S])$

Most Frequently Asked PYQ Topics (Unit 3):

1. Derive the Langmuir isotherm and discuss its limiting cases.
2. State BET assumptions; write BET equation defining all terms; explain how surface area is calculated.
3. Derive the rate of unimolecular surface reaction; discuss order at high and low pressure.

4. Effect of inhibitor on surface reaction kinetics and the energy diagram.
5. Derive Michaelis–Menten equation using the steady-state approximation.
6. Draw and explain the Lineweaver–Burk (double reciprocal) plot; identify slope and intercepts.
7. Effect of pH and temperature on enzyme-catalysed reactions (with graphs).
8. Why is Langmuir better for chemisorption and BET better for physisorption?

Key Analogy: The Langmuir isotherm and Michaelis–Menten equation are *mathematically identical* – both arise from a two-state equilibrium (site empty/filled; enzyme free/bound) with the same hyperbolic form. Recognising this parallel makes both derivations trivial once one is known.

$$\theta = \frac{KP}{1 + KP} \quad \longleftrightarrow \quad v = \frac{V_{\max}[S]}{K_M + [S]}$$